4	**(a)**	period = $2\pi \times 6.7 \times 10^7 / 2.38 \times 10^4 = 17688$ s = 4.91 hours	C1
		period of particle <10 hours 14 minutes, so particle is not in krono-stationary orbit	A1
	(b)	gravitational force on particle provides the centripetal force	B1
		$GMm / r^2 = mv^2 / r$ $M = v^2 r / G = (2.38 \times 10^4)^2 \times 6.7 \times 10^7 / 6.67 \times 10^{-11}$	M1
		$M = 5.7 \times 10^{26}$ kg	A0
	(c)	$GM_T / x^2 = GM_S / (1.2 \times 10^9 - x)^2$	
		gravitational field at point due to Saturn = $GM_S / (1.2 \times 10^9 - x)^2$	C1
		resultant field is zero, so $1.4 \times 10^{23} / x^2 = 5.7 \times 10^{26} / (1.2 \times 10^9 - x)^2$	C1
		$x = 1.9 \times 10^7$ m	A1